UNIT 6 · TOPIC 5

Traction Devices and Tyres — Function, Size and Selection

Syllabus topic: Traction devices: tyre function, size and selection

Traction devices

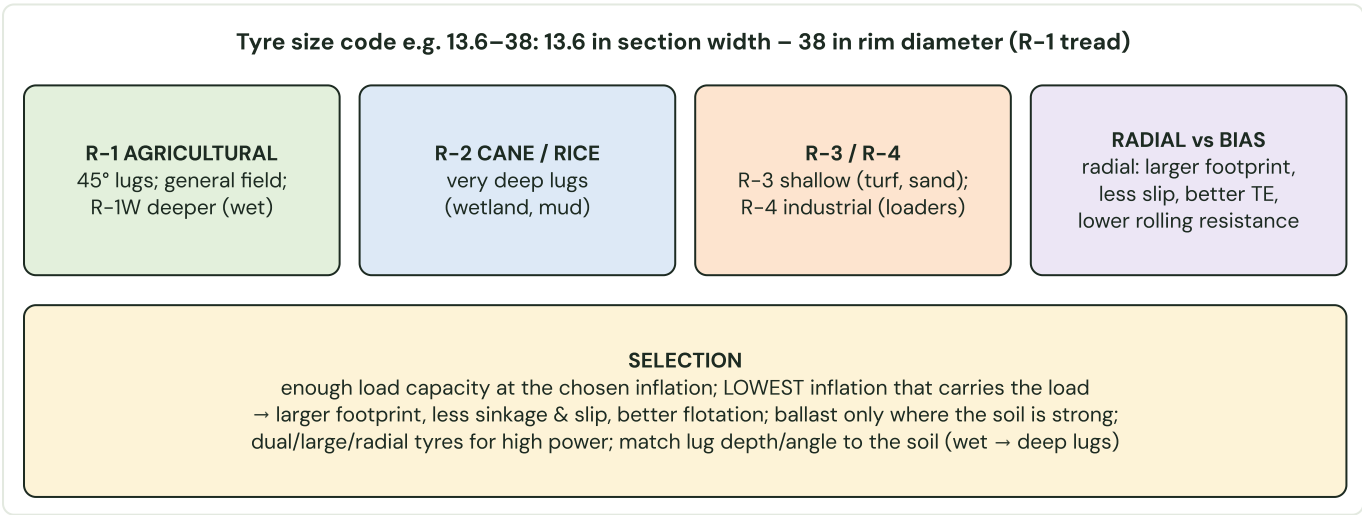
TRACTION = the driving force developed by a wheel, track or other traction device (Liljedahl p. 243).

Device	Use
Pneumatic tyres	The norm
Tracks / half-tracks	Low ground pressure, high traction on soft soil
Cage (lug) wheels, strakes, iron wheels	Puddled paddy (Liljedahl notes lug plates on small paddy machines)

Functions of the pneumatic tyre

- Support the load.
- Develop traction — the lugs shear the soil.
- Cushion the ride.
- Steer.
- All at acceptable ground pressure and wear.

Size and tread classes



Tyre classes and selection (redrawn from the source figure)

- **Size code**, e.g. **13.6–38**: section width 13.6 in, rim diameter 38 in. The ply/load rating and tread class complete the specification.

Tread class	Use
R-1	General agricultural (45° lugs)
R-1W	Deeper lugs, for wet conditions
R-2	Cane/rice — very deep lugs for mud
R-3	Shallow — turf/sand
R-4	Industrial — loaders

RADIAL vs BIAS: radial tyres give a larger, flatter footprint — less slip, lower rolling resistance, better tractive efficiency, fuel saving and longer life (now standard on higher-power tractors).

Selection and ballasting

1. Pick a tyre with adequate load capacity at the intended inflation.
2. Run the LOWEST inflation that carries the load — bigger footprint, higher effective Cn, less sinkage and slip, better flotation and less compaction.
3. Add BALLAST (liquid fill / cast weights) to raise W for pull — but only where the soil is strong. In wet clay ballast sinks and compacts; use cage/flotation wheels instead.
4. Match lug depth/angle to the soil (deep lugs for wet); use dual/large tyres or tracks for high power.
5. Set the correct front–rear weight split (≈ **40/60** in the field, plus draft-control weight transfer — Unit 4, Topics 9a and 10).

Practice MCQs

Q101. In a tyre size code 13.6–38, the number 38 is the:

(a) Section width
(b) Rim (wheel) diameter in inches
(c) Ply rating
(d) Lug angle

Q102. The general-purpose agricultural tyre tread class is:

(a) R-1
(b) R-3
(c) R-4
(d) F-2

Q103. Very deep lugs for wetland/cane-rice tyres denote class:

(a) R-1
(b) R-2
(c) R-3
(d) R-4

Q104. Compared with bias-ply, radial tractor tyres give:

(a) Smaller footprint
(b) Larger footprint, less slip, better TE and wear
(c) More rolling resistance
(d) No difference

Answers

Q101 — (b) Rim (wheel) diameter in inches
Q102 — (a) R-1
Q103 — (b) R-2
Q104 — (b) Larger footprint, less slip, better TE and wear

Sources: Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units, Ch. 10, pp. 240–252; Wismer & Luth (1974), Off-road traction prediction for wheeled vehicles, Trans. ASAE; Freitag (1965) mobility-number dimensional analysis; Bekker pressure–sinkage theory; ASAE S296.2 (slip/rolling radius), S313.2 (cone index); NPTEL traction lectures.